Martin Daigne

Robotics Master's Student — Data Science Minor

martin.daigne@epfl.ch | 06 80 60 18 35 | Lausanne, Switzerland | LinkedIn | GitHub | Website

Education

MSc **École Polytechnique Fédérale de Lausanne (EPFL)**, Robotics - Data Science minor — Switzerland, 2024 – 2027
- Grade: 5.66/6
- Core courses: Machine Learning, Deep Learning, Reinforcement Learning, Convex Optimization, Model Predictive Control, Mobile Robotics, Applied Data Analysis, Modern Natural Language Processing…

Delft University of Technology (TU Delft), Exchange - Mechanical/Aerospace Engineering — Netherlands, 2023 – 2024

BSc **École Polytechnique Fédérale de Lausanne (EPFL)**, Mechanical Engineering — Switzerland, 2021 – 2024
- Grade: 5.57/6
- Core courses: Analysis, General Physics (Mechanics, Thermodynamics, Electromagnetics), Numerical Analysis, Dynamical Systems, Structural Mechanics, Nonlinear Mechanics, Fluid Mechanics, Control Systems, MEMS, Autonomous Vehicles…

Experience

Zeiss Medical Technology, Research Intern — Munich, Germany, Sept 2026 – Feb 2027, 6 months
- Developping new and innovative applications of AI in Ophthalmology.

EPFL, Laboratory of Intelligent Systems (LIS), Semester Project — Switzerland, Feb 2026 – June 2026, 5 months
- Design and implementation of diffusion policies (DDPM, DDIM, Score Matching, Flow Matching) to generate control trajectories for a morphing drone previously controlled with an MPPI.

EPFL, Computational Robot Design and Fabrication Lab (CREATE), Semester Project — Switzerland, Sept 2025 – Jan 2026, 5 months
- Data-Enabled Model Predictive Control for Compliant Manipulators: I built a simulation in Python and an adaptive MPC controller for a soft, tendon-driven arm using CasADi and acados for real-time deployment.

EPFL, Unsteady Flow Diagnostics Laboratory (UNFoLD), Summer in the Lab Internship — Switzerland, June 2024 – Aug 2024, 3 months
- Design, building, and analysis of a random gust generator to improve realism in wind turbine experiments.

Aether Swiss Kite, Control Team Leader — Switzerland, 2025 – 2026, 1 year
- Lead the control team (6 members) in the data acquisition, system identification, and control of the kite to achieve optimized automated control.

Aether Swiss Kite, Electrical and Control Engineer — Switzerland, 2024 – 2025, 1 year
- 1st semester: Design, building, and integration of a position sensor of the kite.
- 2nd semester: Data acquisition and filter implementation (EKF, UKF) to localize the kite.

EPFL Rocket Team, Recovery System Engineer — Switzerland, 2022 – 2023, 1 year
- Design and testing of the parachute of the Nordend rocket of the EPFL Rocket Team for the European EUROC competition.

EPFL, Teaching Assistant — Switzerland
- Teaching assistant in Introduction to Computer Science, Analysis II, Fluid Mechanics, and Model Predictive Control. — 2022–2023 / 2024–2025

Awards

Deep Learning Object Detection Competition — 2025
Winner of the Deep Learning Object Detection competition 2025 in the Image Analysis and Pattern Recognition course at EPFL.

EPFL Space Race — 2022
Winner EPFL Space Race 2022. [link]

Skills

Coding Languages: Strong experience in Python (PyTorch, Pandas, scikit-learn, …). Worked on Matlab and knowledge in C/C++ (mostly for Arduino).

Languages

French: Mother tongue

English: C1 (Cambridge English Advanced Certificate, 2020)

Dutch: Fluent

German: Scholar (Goeth Institut, B1, 2018)